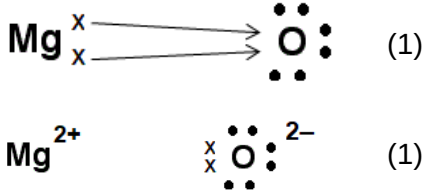


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		 four shared pairs (1) both octets (1)		2		2		
		(ii)		<u>intermolecular</u> forces are weak / forces <u>between</u> molecules are weak (1) accept bonds / interactions / attractions require little energy to overcome / break forces (1) accept doesn't take much heat to overcome / break forces neutral answer - simple molecular do not award any credit if explanation involves covalent bonds	2			2		
	(b)	(i)		 Mg^{2+}  $^{2-}$ (1)		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		(in magnesium oxide) the ions have higher charges (1) electrostatic attraction is greater / attraction between ions is greater / ionic bonds are stronger (1) accept converse for both marks	1	1		2		
		(iii)		88 (3) accept 87.5 if answer incorrect credit each correct step in method $\frac{4.12}{58.5} = 0.0704$ (1) $\frac{0.0704}{0.080} = 0.88$ (1) $0.88 \times 100 = 88$ (1) alternative method $0.080 \times 58.5 / 4.68$ (1) $\frac{4.12}{4.68} = 0.88$ (1) $0.88 \times 100 = 88$ (1)		3		3	3	
				Question 5 total	3	8	0	11	3	0